

Choose Rock for the Job

BUILD • Science • 40 minutes

I can choose a supplied rock sample for a stated model job and justify the decision with relevant property evidence and a limit.

Prepare before pupils arrive

Setup: Prepare a low-density matrix and two-candidate version for each brief. Keep all contexts explicitly fictional and classroom-scale.

Safety: Do not present the decision as real construction advice. A complete no-touch/director route is available for every tier.

Equipment: Fictional job briefs, candidate evidence matrix, decision sheet, audience question card and pencil or communication aid.

Safety: Use cards or sealed lightweight display samples only. No real construction, heavy lifting, scraping, wet testing, breaking, climbing or site work is required.

Fallback: Use two candidate cards and one enlarged brief at a time. An adult scribes exact pupil reasoning and the audience response.

Access and participation

Offer reading aloud, pointing, signing, quiet thinking, a no-touch route and exact scribing where these support the learner. Keep the same subject decision. In shared work, let each learner commit before feedback; use a partner as card mover or checker, then swap. Avoid public speed rankings.

Source lesson curriculum link: Identify everyday uses of different rocks (links D&T / Construction).

40-minute teaching sequence

Stage / total time	Teaching move	Adult support / response
Arrival · 3 min	Last lesson: give one job-property because link. Last lesson: why could a drainage job value water movement?	Wait, regulate and retrieve through the lightest workable route. If recall is inaccessible, point to one retrieval sentence; do not teach today's answer.
Starter · 3 min	Think first. Point, say, sign or write your choice.	Protect curiosity: receive every first idea without judgement. Ask 'What makes you think that?' and preserve the prediction for later evidence.
I do · 4 min	criteria — the job rules candidate — one possible choice justify — give evidence for why Underline the criteria and circle only the evidence columns that answer them. Ignore colour unless appearance is named in the brief.	Let the teacher/model carry the explanation; pupils watch, point or track. Use a visual cue only if access is the barrier, then fade it.

Stage / total time	Teaching move	Adult support / response
We do · 4 min	Commit to an answer. Show the evidence you used.	Scan every response mode. Do not infer understanding from handwriting or confidence. Secure: transfer. Mixed: contrast. Misconception: counterexample then a fresh question.
I do 2 · 3 min	Add one limit before the audience responds: the evidence describes supplied samples in classroom tests, not every specimen or real installation. Invite a question about criteria, evidence or a missing check.	Model one complete evidence-to-explanation sentence, then pause. Keep the science words; change density, modality or adult role before changing the goal.
We do 2 · 4 min	Route each candidate evidence card to RAIN COVER, PATH, INDOOR DISPLAY or NEEDS ANOTHER CHECK. Read the brief criterion.	Protect pupil operation and thinking; support handling only where needed. Ask what changed and what stayed the same before supplying a scientific word.
Independent · 15 min	Choose a supplied candidate for one model job, compare at least two options and revise or confirm the decision after a genuine audience question. Record: Brief criteria, two-candidate comparison, chosen candidate, relevant because evidence, one limit, one audience question and visible response.	Protect independence. Accept equivalent evidence routes and scribe exact meaning. Prompt only as much as needed; fade as soon as the pupil continues.
Exit · 4 min	State your choice, strongest evidence and one limit after the audience response. Point to, read or explain the evidence you made.	Genuinely receive one final response and name back the Science heard or seen. Choose one visible next move; written closure is optional when Voice and Audience are observable.

The timing belongs to the whole stage. Several PowerPoint slides may share it; do not restart that time on every slide. Full source-stage text is in the PowerPoint speaker notes and original HTML.

Answers, evidence and feedback

Hinge answer: Choose P because it had a faint scratch mark and the surface card shows a stable, even path piece; keep grip as a separate safety check.

Correct. It uses wear and surface evidence that answer the path brief and keeps the untested safety check open.

Misconception repair

Repaired. A justified choice follows the brief; irrelevant evidence cannot become points just to force a winner.

Guided checkpoint

Correct. It uses wear and surface evidence that answer the path brief and keeps the untested safety check open. Return to the brief: repeated contact and surface suitability matter more here than colour or roof-style water control.

Candidate R

RAIN-COVER MODEL. Rain-cover candidate: R directly answers the water and overlap criteria in this model brief. Which brief needs low passage and flat overlapping pieces?

R caution

NEEDS ANOTHER CHECK. Needs another check outside the model: the classroom result has a time and scale limit. This statement controls how far the choice may travel.

Candidate P

HARD-WEARING PATH MODEL. Path candidate: P answers the supplied wear and surface criteria. Which brief focuses on repeated surface contact?

P safety gap

NEEDS ANOTHER CHECK. Needs another check: the matrix does not decide real path safety. This card names evidence the classroom tests did not collect.

Candidate D

INDOOR DISPLAY. Display candidate: appearance is relevant because the brief is visual and indoors. For which brief is pattern named as a criterion rather than used to guess another property?

Colour-only vote

NEEDS ANOTHER CHECK. Needs another check: preference does not answer the wear and safety criteria. Return to what the path must do, not which sample is most liked.

Candidate U

NEEDS ANOTHER CHECK. Needs another check: the candidate lacks relevant evidence for any stated job. A photograph alone cannot fill the evidence matrix.

Display stability

INDOOR DISPLAY. Display match: it preserves the stated indoor and stability conditions rather than claiming D suits everything. This evidence belongs with the brief whose criteria include appearance and indoor use.

Model limitation

The fixed matrix is fictional and simplified. It supports classroom reasoning only; it cannot approve real materials, structures, costs, installation or safety.

Independent evidence

Brief criteria, two-candidate comparison, chosen candidate, relevant because evidence, one limit, one audience question and visible response.

Supported: Choose between two candidates for one brief and complete an oral, pointed or scribed because sentence plus one question-mark limit. Use: The job needs ___. I choose ___ because ___. We still need to check ___.

Standard: Compare three candidates on relevant criteria, justify one choice and revise or confirm it after an audience question. Cross out irrelevant columns before comparing.

Stretch: Evaluate a trade-off, defend why one tempting candidate is unsuitable and explain how the decision would change for a different purpose. Keep classroom evidence separate from real construction approval.

Record the teaching response

What the learner actually showed	Next teaching action
Secure explanation with evidence	Reduce content prompting; offer another example or a limitation.
Partly correct / mixed	Point to the deciding evidence and invite a revision.
Access barrier	Change the reading, sensory or response format; keep the subject decision.
Missing source or observation	Keep the gap visible. Name the source or authorised check needed.

Safety and evidence boundary

Use the centre's authorised practical or fieldwork method and local risk assessment. Supplied sample data, fictional place models, card feedback and rehearsals are teaching material. They are not the learner's practical observations or proof of a qualification outcome. For portfolio use, check the exact registered unit/version, accepted evidence and centre process.

Sources and companion notes

Primary classroom source: SCI_B_W12B_Choose_Rock_For_The_Job_Do.html

The uploaded lesson is included unchanged. Text and teaching steps in the PowerPoint are editable; source diagrams are placed images. Word booklets are editable. Static PDFs cannot reproduce HTML controls; use the paper cards/tables or open the original HTML.

Verification scope: matched to the uploaded title, pathway, objective, stage sequence and 40-minute timing. Embedded workbook references are retained as source locators; this is not a new line-by-line audit of every scheme-of-work row or a centre assessment sign-off.